

Course 5083C

5 Credits: 4

Program Elective

Source-grounded

Robotics and Control

□ To understand concepts, parts of robots and types of robots.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5083C is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5083C.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Instrumentation Engineering
Course code	5083C
Course title	Robotics and Control
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:3 T:1 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

13 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ To understand concepts, parts of robots and types of robots.
- □ To familiarize with the various drive systems for robot, sensors and their applications in robotic automation.
- □ To discuss about the various applications of robots, justification and implementation
- □ To conceptualize automation and understand applications of robots in various

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding characteristics of robot. Explain the various robotic actuators on hydraulic,	See official syllabus
CO2	17 Applying pneumatic and electrical drives. Explain about various types of sensors and robot	See official syllabus
CO3	16 Understanding vision system.	See official syllabus
CO4	Explain the various applications of robots. 10 Understanding	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 3
CO3	CO3 3
CO4	CO4 2

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	advantages and disadvantages.	4	As specified
Module 2	electrical drives.	3	As specified
Module 3	system.	3	As specified
Module 4	Explain the various applications of robots.	3	As specified

MODULE 1

advantages and disadvantages.

This module is presented from the official syllabus scope for Robotics and Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: advantages and disadvantages.
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

2 Understanding laws of robotics.

2 Understanding laws of robotics. is an official topic in Module 1 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding laws of robotics. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding laws of robotics. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding laws of robotics. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-2 Understanding laws of robotics.

രൂപതരതതതതതതതതതതതതതത definition/meaning തതതതതതതതതതതതതത. തതതതതത തതതതതത തതതതതതതത, steps, application തതതതതത തതതതതതതതതത തതതതതത. Technical terms English- തതതതത തതതതതതതതതതതത.

Describe the Robot anatomy

Describe the Robot anatomy is an official topic in Module 1 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the Robot anatomy using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the robot anatomy should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the Robot anatomy means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-റോബോട്ടിക്സ് Describe the Robot anatomy റോബോട്ടിക്സ് റോബോട്ടിക്സ് definition/meaning റോബോട്ടിക്സ് റോബോട്ടിക്സ്. റോബോട്ടിക്സ് റോബോട്ടിക്സ് steps, application റോബോട്ടിക്സ് റോബോട്ടിക്സ്. Technical terms English-റോബോട്ടിക്സ് റോബോട്ടിക്സ്.

Explain the classification of robots 6 Understanding List the advantages and disadvantages of

Explain the classification of robots 6 Understanding List the advantages and disadvantages of is an official topic in Module 1 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the classification of robots 6 Understanding List the advantages and disadvantages of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the classification of robots 6 understanding list the advantages and disadvantages of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the classification of robots 6 Understanding List the advantages and disadvantages of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-റോബോട്ട് Explain the classification of robots 6 Understanding List the advantages and disadvantages of റോബോട്ട് definition/meaning റോബോട്ട്. റോബോട്ട് റോബോട്ട് റോബോട്ട്, steps, application റോബോട്ട് റോബോട്ട് റോബോട്ട്. Technical terms English-റോബോട്ട് റോബോട്ട്.

2 Understanding robots Fundamentals of Robotics: Introduction; Degree of freedom, Asimov's laws of robotics, Definition; Robot anatomy (parts) and its working; Robot Components: Manipulator, End effectors; Construction of links, Types of joints; Classification of robots; Cartesian, Cylindrical, Spherical, Scara, Vertical articulated; Robot work Volumes, Comparison; Advantages and disadvantages of robots. Explain the various robotic actuators on hydraulic, pneumatic and

2 Understanding robots Fundamentals of Robotics: Introduction; Degree of freedom, Asimov's laws of robotics, Definition; Robot anatomy (parts) and its working; Robot Components: Manipulator, End effectors; Construction of links, Types of joints; Classification of robots; Cartesian, Cylindrical, Spherical, Scara, Vertical articulated; Robot work Volumes, Comparison; Advantages and disadvantages of robots. Explain the various robotic actuators on hydraulic, pneumatic and is an official topic in Module 1 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding robots Fundamentals of Robotics: Introduction; Degree of freedom, Asimov's laws of robotics, Definition; Robot anatomy (parts) and its working; Robot Components: Manipulator, End effectors; Construction of links, Types of joints; Classification of robots; Cartesian, Cylindrical, Spherical, Scara, Vertical articulated; Robot work Volumes, Comparison; Advantages and disadvantages of robots. Explain the various robotic actuators on hydraulic, pneumatic and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding robots fundamentals of robotics: introduction; degree of freedom, asimov's laws of robotics, definition; robot anatomy (parts) and its

working; robot components: manipulator, end effectors; construction of links, types of joints; classification of robots; cartesian, cylindrical, spherical, scara, vertical articulated; robot work volumes, comparison; advantages and disadvantages of robots. explain the various robotic actuators on hydraulic, pneumatic and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding robots Fundamentals of Robotics: Introduction; Degree of freedom, Asimov's laws of robotics, Definition; Robot anatomy (parts) and its working; Robot Components: Manipulator, End effectors; Construction of links, Types of joints; Classification of robots; Cartesian, Cylindrical, Spherical, Scara, Vertical articulated; Robot work Volumes, Comparison; Advantages and disadvantages of robots. Explain the various robotic actuators on hydraulic, pneumatic and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-2 Understanding robots Fundamentals of Robotics: Introduction; Degree of freedom, Asimov's laws of robotics, Definition; Robot anatomy (parts) and its working; Robot Components: Manipulator, End effectors; Construction of links, Types of joints; Classification of robots; Cartesian, Cylindrical, Spherical, Scara, Vertical articulated; Robot work Volumes, Comparison; Advantages and disadvantages of robots. Explain the various robotic actuators on hydraulic, pneumatic and definition/meaning. steps, application. Technical terms English-.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

advantages and disadvantages.

M1.01	Explain the fundamentals of Robotics & laws of robotics.	2
Understanding		
M1.02	Describe the Robot anatomy	5
Understanding		
M1.03	Explain the classification of robots	6
Understanding		
M1.04	List the advantages and disadvantages of robots	2
Understanding		
Contents:		
Fundamentals of Robotics: Introduction; Degree of freedom, Asimov's laws of robotics,		
Definition; Robot anatomy (parts) and its working; Robot Components: Manipulator, End		
effectors; Construction of links, Types of joints; Classification of robots; Cartesian,		
Cylindrical, Spherical, Scara, Vertical articulated; Robot work Volumes, Comparison;		
Advantages and disadvantages of robots.		
Explain the various robotic actuators on hydraulic, pneumatic and		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

electrical drives.

This module is presented from the official syllabus scope for Robotics and Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: electrical drives.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain different Robotic Drive Systems 5 Understanding Explain different actuation methods 4 Understanding M2.02 Conversion between linear and rotary

Explain different Robotic Drive Systems 5 Understanding Explain different actuation methods 4 Understanding M2.02 Conversion between linear and rotary is an official topic in Module 2 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different Robotic Drive Systems 5 Understanding Explain different actuation methods 4 Understanding M2.02 Conversion between linear and rotary using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different robotic drive systems 5 understanding explain different actuation methods 4 understanding m2.02 conversion between linear and rotary should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different Robotic Drive Systems 5 Understanding Explain different actuation methods 4 Understanding M2.02 Conversion between linear and rotary means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നു വിശദീകരിക്കുക. Explain different Robotic Drive Systems 5 Understanding Explain different actuation methods 4 Understanding M2.02 Conversion between linear and rotary ഡിഫിനിഷൻ/മനസ്സിലാക്കൽ definition/meaning ഡിഫിനിഷൻ/മനസ്സിലാക്കൽ. ഡിഫിനിഷൻ/മനസ്സിലാക്കൽ, steps, application ഡിഫിനിഷൻ/മനസ്സിലാക്കൽ. Technical terms English-എന്നു വിശദീകരിക്കുക.

4 Applying motion

4 Applying motion is an official topic in Module 2 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying motion using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying motion should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying motion means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-4 Applying motion പരമ്പരപ്പെടുത്തൽ പരമ്പര
definition/meaning പരമ്പരപ്പെടുത്തൽ. പരമ്പര പരമ്പര പരമ്പര, steps, application
പരമ്പര പരമ്പരപ്പെടുത്തൽ പരമ്പര. Technical terms English-4 പരമ്പര പരമ്പരപ്പെടുത്തൽ.

Explain the feedback devices 4 Understanding Robotic Drive System: Actuators; Hydraulic, Pneumatic and Electrical drives; Linear actuator; Rotary drives; AC servo motor; DC servo motors and Stepper motors; Conversion between linear and rotary motion; Feedback devices; Potentiometers; Optical encoders; DC tachometers; Explain about various types of sensors and concepts on robot vision

Explain the feedback devices 4 Understanding Robotic Drive System: Actuators; Hydraulic, Pneumatic and Electrical drives; Linear actuator; Rotary drives; AC servo motor; DC servo motors and Stepper motors; Conversion between linear and rotary motion; Feedback devices; Potentiometers; Optical encoders; DC tachometers; Explain about various types of sensors and concepts on robot vision is an official topic in Module 2 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the feedback devices 4 Understanding Robotic Drive System: Actuators; Hydraulic, Pneumatic and Electrical drives; Linear actuator; Rotary drives; AC servo motor; DC servo motors and Stepper motors; Conversion between linear and rotary motion; Feedback devices; Potentiometers; Optical encoders; DC tachometers; Explain about various types of sensors and concepts on robot vision using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the feedback devices 4 understanding robotic drive system: actuators; hydraulic, pneumatic and electrical drives; linear actuator; rotary drives; ac servo motor; dc servo motors and stepper motors; conversion between linear and rotary motion; feedback devices; potentiometers; optical encoders; dc tachometers; explain about various types of sensors and concepts on robot vision

should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the feedback devices 4 Understanding Robotic Drive System: Actuators; Hydraulic, Pneumatic and Electrical drives; Linear actuator; Rotary drives; AC servo motor; DC servo motors and Stepper motors; Conversion between linear and rotary motion; Feedback devices; Potentiometers; Optical encoders; DC tachometers; Explain about various types of sensors and concepts on robot vision means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain the feedback devices 4 Understanding Robotic Drive System: Actuators; Hydraulic, Pneumatic and Electrical drives; Linear actuator; Rotary drives; AC servo motor; DC servo motors and Stepper motors; Conversion between linear and rotary motion; Feedback devices; Potentiometers; Optical encoders; DC tachometers; Explain about various types of sensors and concepts on robot vision definition/meaning .
 , steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

electrical drives.

M2.01	Explain different Robotic Drive Systems	5	
Understanding			
	Explain different actuation methods	4	
Understanding			
M2.02	Conversion between linear and rotary		
M2.03	motion	4	Applying
M2.04	Explain the feedback devices	4	
Understanding			
	Series Test - I	1	

Contents:

Robotic Drive System: Actuators; Hydraulic, Pneumatic and Electrical drives; Linear actuator; Rotary drives; AC servo motor; DC servo motors and Stepper motors; Conversion between linear and rotary motion; Feedback devices; Potentiometers; Optical encoders; DC tachometers;

Explain about various types of sensors and concepts on robot vision

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

system.

This module is presented from the official syllabus scope for Robotics and Control.
Use the topic cards for learning and the source transcript for exact coverage.

As specified

CO-linked

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: system.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain the requirements of sensors 4 Understanding Explain the application of different

Explain the requirements of sensors 4 Understanding Explain the application of different is an official topic in Module 3 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the requirements of sensors 4 Understanding Explain the application of different using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the requirements of sensors 4 understanding explain the application of different should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the requirements of sensors 4 Understanding Explain the application of different means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
Explain the requirements of sensors 4
Understanding Explain the application of different
definition/meaning
definition/meaning steps, application
definition/meaning Technical terms English-
definition/meaning

sensors Explain the robot's machine vision

sensors Explain the robot's machine vision is an official topic in Module 3 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify sensors Explain the robot's machine vision using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, sensors explain the robot's machine vision should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What sensors Explain the robot's machine vision means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-സെൻസറുകൾ Explain the robot's machine vision സെൻസറുകൾ definition/meaning സെൻസറുകൾ. സെൻസറുകൾ സെൻസറുകൾ, steps, application സെൻസറുകൾ സെൻസറുകൾ. Technical terms English-സെൻസറുകൾ സെൻസറുകൾ.

system Applications of robot vision systems. 3 Understanding M3.04 Sensors: Requirements of a sensor; Principles and Applications of the following types of sensors: Position sensors (Encoders, Resolvers, Piezo Electric); Range sensors; Proximity sensing; Force and torque sensing, Tactile sensors. Introduction to Machine Vision: Robot vision system (scanning and digitizing image data); Cameras (Acquisition of images); Vidicon camera (Working principle & construction); Applications of Robot vision system: Inspection, Identification, Navigation & serving.

system Applications of robot vision systems. 3 Understanding M3.04 Sensors: Requirements of a sensor; Principles and Applications of the following types of sensors: Position sensors (Encoders, Resolvers, Piezo Electric); Range sensors; Proximity sensing; Force and torque sensing, Tactile sensors. Introduction to Machine Vision: Robot vision system (scanning and digitizing image data); Cameras (Acquisition of images); Vidicon camera (Working principle & construction); Applications of Robot vision system: Inspection, Identification, Navigation & serving. is an official topic in Module 3 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify system Applications of robot vision systems. 3 Understanding M3.04 Sensors: Requirements of a sensor; Principles and Applications of the following types of sensors: Position sensors (Encoders, Resolvers, Piezo Electric); Range sensors; Proximity sensing; Force and torque sensing, Tactile sensors. Introduction to Machine Vision: Robot vision system (scanning and digitizing image data); Cameras (Acquisition of images); Vidicon camera (Working principle & construction); Applications of Robot vision system: Inspection, Identification, Navigation & serving. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, system applications of robot vision systems. 3 understanding m3.04 sensors: requirements of a sensor; principles and applications of the following types of sensors: position sensors (encoders, resolvers, piezo electric); range sensors; proximity sensing; force and torque sensing, tactile sensors. introduction to machine vision: robot vision system (scanning and digitizing image data); cameras (acquisition of images); vidicon camera (working principle & construction); applications of robot vision system: inspection, identification, navigation & serving. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What system Applications of robot vision systems. 3 Understanding M3.04 Sensors: Requirements of a sensor; Principles and Applications of the following types of sensors: Position sensors (Encoders, Resolvers, Piezo Electric); Range sensors; Proximity sensing; Force and torque sensing, Tactile sensors. Introduction to Machine Vision: Robot vision system (scanning and digitizing image data); Cameras (Acquisition of images); Vidicon camera (Working principle & construction); Applications of Robot vision system: Inspection, Identification, Navigation & serving. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-റോബറ്റ് സിസ്റ്റം Applications of robot vision systems. 3 Understanding M3.04 Sensors: Requirements of a sensor; Principles and Applications of the following types of sensors: Position sensors (Encoders, Resolvers, Piezo Electric); Range sensors; Proximity sensing; Force and torque sensing, Tactile sensors. Introduction to Machine Vision: Robot vision system (scanning and digitizing image data); Cameras (Acquisition of images); Vidicon camera (Working principle & construction); Applications of Robot vision system: Inspection, Identification, Navigation & serving. റോബറ്റ് വിഷ്ണുവിന്റെ നിർവ്വചനം/അർത്ഥം റോബറ്റ് വിഷ്ണുവിന്റെ ഘടന, steps, application റോബറ്റ് വിഷ്ണുവിന്റെ ഉപയോഗം. Technical terms English-റോബറ്റ് വിഷ്ണുവിന്റെ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

system.

M3.01	Explain the requirements of sensors	4
Understanding	Explain the application of different	4
Understanding		
M3.02	sensors	
	Explain the robot's machine vision	5
Understanding		
M3.03	system	
	Applications of robot vision systems.	3
Understanding		
M3.04		
Contents:		
Sensors:	Requirements of a sensor; Principles and Applications of the following types of	
sensors:	Position sensors (Encoders, Resolvers, Piezo Electric); Range sensors; Proximity	
sensing;	Force and torque sensing, Tactile sensors. Introduction to Machine Vision: Robot vision system (scanning and digitizing image data); Cameras (Acquisition of images); Vidicon camera (Working principle & construction); Applications of Robot vision system: Inspection, Identification, Navigation & serving.	

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Explain the various applications of robots.

This module is presented from the official syllabus scope for Robotics and Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified

CO-linked

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the various applications of robots.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

4 Understanding system Explain the advanced automation

4 Understanding system Explain the advanced automation is an official topic in Module 4 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding system Explain the advanced automation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding system explain the advanced automation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding system Explain the advanced automation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4- 4 Understanding system Explain the advanced automation ഓട്ടോമേറ്റേഷൻ definition/meaning ഓട്ടോമേറ്റേഷൻ. ഓട്ടോമേറ്റേഷൻ ഓട്ടോമേറ്റേഷൻ, steps, application ഓട്ടോമേറ്റേഷൻ ഓട്ടോമേറ്റേഷൻ. Technical terms English- ഓട്ടോമേറ്റേഷൻ ഓട്ടോമേറ്റേഷൻ.

functions Explain the applications of robotics in

functions Explain the applications of robotics in is an official topic in Module 4 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify functions Explain the applications of robotics in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, functions explain the applications of robotics in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What functions Explain the applications of robotics in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-റോബോട്ടിക്സ് functions Explain the applications of robotics in different fields. റോബോട്ടിക്സ് definition/meaning റോബോട്ടിക്സ് റോബോട്ടിക്സ്. റോബോട്ടിക്സ് റോബോട്ടിക്സ് റോബോട്ടിക്സ്, steps, application റോബോട്ടിക്സ് റോബോട്ടിക്സ് റോബോട്ടിക്സ്. Technical terms English-റോബോട്ടിക്സ് റോബോട്ടിക്സ്.

3 Understanding various industries. Automation: Basic elements of automated system, advanced automation functions, levels of automation. Industrial Applications: Application of robots in machining; welding; assembly, material handling, biomedical applications. Text / Reference: T/R Book Title/Author Mikell P. Weiss G.M., Nagel R.N., Odraj N.G., Industrial Robotics, McGraw R1 Hill Singapore, 1996 Ghosh, Control in Robotics and Automation: Sensor Based Integration, Allied R2 Publishers, Chennai, 1998. Saeed B. Niku, - Introduction to Robotics: Analysis, Systems, Applications - R3 Pearson Education Inc. New Delhi 2006. M.P. Groover, - Industrial Robotics: Technology, Programming and R4 Applications -Tata McGraw Hill Co, 2001. Fu.K.S. Gonzalz.R.C and Lee C.S.G - Robotics Control, Sensing, Vision and R5 Intelligence -, McGraw Hill Book Co, 1987. R6 Yoram Koren - Robotics for Engineers -McGraw Hill Book Co, 1992. Ganesh S. Hedge, - A Text book on Industrial Robotics -Laxmi Publications R7 Pvt. Ltd., New Delhi, 2008. S.R. Deb & Sankha Deb - Robotics Technology and Flexible Automation - R8 Tata McGraw-Hill, 2010. Elements of Robotics Process Automation, Mukherjee, Khanna Publishing R9 House, Delhi, 2018 R10 Issac Asimov I Robot, Ballantine Books, New York, 1986. Thomas R. Kurfess Ph.D., P.E. Robotics And Automation Handbook CRC R11 press Online Resources: Sl.No Website Link 1 <https://nptel.ac.in/courses/112/101/112101098/> 2 <http://www.nptelvideos.in/2012/12/robotics.html> 3 <https://www.guru99.com/robotic-process-automation-tutorial.html>

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<https://nptel.ac.in/courses/112/101/112101098/> 2

<http://www.nptelvideos.in/2012/12/robotics.html> 3

<https://www.guru99.com/robotic-process-automation-tutorial.html> is an official topic in Module 4 of Robotics and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding various industries. Automation: Basic elements of automated system, advanced automation functions, levels of automation. Industrial Applications: Application of robots in machining; welding; assembly, material handling, biomedical applications. Text / Reference: T/R Book Title/Author Mikell P. Weiss G.M., Nagel R.N., Odraj N.G., Industrial Robotics, McGraw R1 Hill Singapore, 1996 Ghosh, Control in Robotics and Automation: Sensor Based Integration, Allied R2 Publishers, Chennai, 1998. Saeed B. Niku, - Introduction to Robotics: Analysis, Systems, Applications – R3 Pearson Education Inc. New Delhi 2006. M.P. Groover, - Industrial Robotics: Technology, Programming and R4 Applications –Tata McGraw Hill Co, 2001. Fu.K.S. Gonzalz.R.C and Lee C.S.G - Robotics Control, Sensing, Vision and R5 Intelligence –, McGraw Hill Book Co, 1987. R6 Yoram Koren - Robotics for Engineers –McGraw Hill Book Co, 1992. Ganesh S. Hedge, - A Text book on Industrial Robotics –Laxmi Publications R7 Pvt. Ltd., New Delhi, 2008. S.R. Deb & Sankha Deb - Robotics Technology and Flexible Automation – R8 Tata McGraw-Hill, 2010. Elements of Robotics Process Automation, Mukherjee, Khanna Publishing R9 House, Delhi, 2018 R10 Issac Asimov I Robot, Ballantine Books, New York, 1986. Thomas R. Kurfess Ph.D., P.E. Robotics And Automation Handbook CRC R11 press Online Resources: Sl.No Website Link 1
<https://nptel.ac.in/courses/112/101/112101098/> 2
<http://www.nptelvideos.in/2012/12/robotics.html> 3
<https://www.guru99.com/robotic-process-automation-tutorial.html> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding various industries. automation: basic elements of automated system, advanced automation functions, levels of automation. industrial applications: application of robots in machining; welding; assembly, material handling, biomedical applications. text / reference: t/r book title/author mikell p. weiss g.m., nagel r.n., odraj n.g., industrial robotics, mcgraw r1 hill singapore, 1996 ghosh, control in robotics and automation: sensor based integration, allied r2 publishers, chennai, 1998. saeed b. niku, - introduction to robotics: analysis, systems, applications - r3 pearson education inc. new delhi 2006. m.p. groover, - industrial robotics: technology, programming and r4 applications -tata mcgraw hill co, 2001. fu.k.s. gonzalz.r.c and lee c.s.g - robotics control, sensing, vision and r5 intelligence -, mcgraw hill book co, 1987. r6 yoram koren - robotics for engineers -mcgraw hill book co, 1992. ganesh s. hedge, - a text book on industrial robotics -laxmi publications r7 pvt. ltd., new delhi, 2008. s.r. deb & sankha deb - robotics technology and flexible automation - r8 tata mcgraw-hill, 2010. elements of robotics process automation, mukherjee, khanna publishing r9 house, delhi, 2018 r10 issac asimov i robot, ballantine books, new york, 1986. thomas r. kurfess ph.d., p.e. robotics and automation handbook crc r11 press online resources: sl.no website link 1 <https://nptel.ac.in/courses/112/101/112101098/> 2 <http://www.nptelvideos.in/2012/12/robotics.html> 3 <https://www.guru99.com/robotic-process-automation-tutorial.html> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding various industries. Automation: Basic elements of automated system, advanced automation functions, levels of automation. Industrial Applications: Application of robots in machining; welding; assembly, material handling, biomedical applications. Text / Reference: T/R Book Title/Author Mikell P. Weiss G.M., Nagel R.N., Odraj N.G., Industrial Robotics, McGraw R1 Hill Singapore, 1996 Ghosh, Control in Robotics and Automation: Sensor Based Integration, Allied R2 Publishers, Chennai, 1998. Saeed B. Niku, - Introduction to Robotics: Analysis, Systems, Applications - R3 Pearson Education Inc. New Delhi 2006. M.P. Groover, - Industrial Robotics: Technology, Programming and R4 Applications -Tata McGraw Hill Co, 2001. Fu.K.S. Gonzalz.R.C and Lee C.S.G - Robotics Control, Sensing, Vision and R5 Intelligence -, McGraw Hill Book Co, 1987. R6 Yoram Koren - Robotics for Engineers -McGraw Hill Book Co, 1992. Ganesh S. Hedge, - A Text book on Industrial Robotics -Laxmi Publications R7 Pvt. Ltd., New Delhi, 2008. S.R. Deb & Sankha Deb - Robotics Technology and Flexible Automation - R8 Tata McGraw-Hill, 2010. Elements of Robotics Process Automation, Mukherjee, Khanna Publishing R9 House, Delhi, 2018 R10 Issac Asimov I Robot, Ballantine Books, New York, 1986. Thomas R. Kurfess Ph.D., P.E. Robotics And Automation Handbook CRC R11 press Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/112/101/112101098/ 2 http://www.nptelvideos.in/2012/12/robotics.html 3 https://www.guru99.com/robotic-process-automation-tutorial.html means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 3 Understanding various industries.

Automation: Basic elements of automated system, advanced automation functions, levels of automation. Industrial Applications: Application of robots in machining; welding; assembly, material handling, biomedical applications. Text / Reference: T/R Book Title/Author Mikell P. Weiss G.M., Nagel R.N., Odraj N.G., Industrial Robotics, McGraw R1 Hill Singapore, 1996 Ghosh, Control in Robotics and Automation: Sensor Based Integration, Allied R2 Publishers, Chennai, 1998. Saeed B. Niku, - Introduction to Robotics: Analysis, Systems, Applications – R3 Pearson Education Inc. New Delhi 2006. M.P. Groover, - Industrial Robotics: Technology, Programming and R4 Applications –Tata McGraw Hill Co, 2001. Fu.K.S. Gonzalz.R.C and Lee C.S.G - Robotics Control, Sensing, Vision and R5 Intelligence –, McGraw Hill Book Co, 1987. R6 Yoram Koren - Robotics for Engineers –McGraw Hill Book Co, 1992. Ganesh S. Hedge, - A Text book on Industrial Robotics –Laxmi Publications R7 Pvt. Ltd., New Delhi, 2008. S.R. Deb & Sankha Deb - Robotics Technology and Flexible Automation – R8 Tata McGraw-Hill, 2010. Elements of Robotics Process Automation, Mukherjee, Khanna Publishing R9 House, Delhi, 2018 R10 Issac Asimov I Robot, Ballantine Books, New York, 1986. Thomas R. Kurfess Ph.D., P.E. Robotics And Automation Handbook CRC R11 press Online Resources: Sl.No Website Link 1 <https://nptel.ac.in/courses/112/101/112101098/> 2 <http://www.nptelvideos.in/2012/12/robotics.html> 3 <https://www.guru99.com/robotic-process-automation-tutorial.html> 4 definition/meaning 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 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Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Explain the various applications of robots.

Describe the elements of automated

M4.01

4

Understanding

system

Explain the advanced automation

3

Understanding

M4.02

functions

Explain the applications of robotics in

M4.03

3

Understanding

various industries.

Series Test - II

1

Contents:

Automation: Basic elements of automated system, advanced automation functions, levels

of automation. Industrial Applications: Application of robots in machining; welding;

assembly, material handling, biomedical applications.

Text / Reference:

T/R

Book Title/Author

Mikell P. Weiss G.M., Nagel R.N., Odraj N.G., Industrial Robotics,

McGraw

R1

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Understanding laws of robotics.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding laws of robotics..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Describe the Robot anatomy. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the Robot anatomy*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Explain the classification of robots 6 Understanding List the advantages and disadvantages of. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the classification of robots 6 Understanding List the advantages and disadvantages of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 2 Understanding robots Fundamentals of Robotics: Introduction; Degree of freedom, Asimov's laws of robotics, Definition; Robot anatomy (parts) and its working; Robot Components: Manipulator, End effectors; Construction of links, Types of joints; Classification of robots; Cartesian, Cylindrical, Spherical, Scara, Vertical articulated; Robot work Volumes, Comparison; Advantages and disadvantages of robots. Explain the various robotic actuators on hydraulic, pneumatic and. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding robots Fundamentals of Robotics: Introduction; Degree of freedom, Asimov's laws of robotics, Definition; Robot anatomy (parts) and its working; Robot Components: Manipulator, End effectors; Construction of links, Types of joints; Classification of robots; Cartesian, Cylindrical, Spherical, Scara, Vertical articulated; Robot work Volumes, Comparison;*

Advantages and disadvantages of robots. Explain the various robotic actuators on hydraulic, pneumatic and. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain Explain different Robotic Drive Systems 5 Understanding Explain different actuation methods 4 Understanding M2.02 Conversion between linear and rotary. (3 marks)

Answer: Begin with the standard definition or identity of *Explain different Robotic Drive Systems 5 Understanding Explain different actuation methods 4 Understanding M2.02 Conversion between linear and rotary*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Applying motion. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying motion*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain the feedback devices 4 Understanding Robotic Drive System: Actuators; Hydraulic, Pneumatic and Electrical drives; Linear actuator; Rotary drives; AC servo motor; DC servo motors and Stepper motors; Conversion between linear and rotary motion; Feedback devices;

Potentiometers; Optical encoders; DC tachometers; Explain about various types of sensors and concepts on robot vision. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the feedback devices 4 Understanding Robotic Drive System: Actuators; Hydraulic, Pneumatic and Electrical drives; Linear actuator; Rotary drives; AC servo motor; DC servo motors and Stepper motors; Conversion between linear and rotary motion; Feedback devices; Potentiometers; Optical encoders; DC tachometers; Explain about various types of sensors and concepts on robot vision.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Explain the requirements of sensors 4 Understanding Explain the application of different. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the requirements of sensors 4 Understanding Explain the application of different.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain sensors Explain the robot's machine vision. (3 marks)

Answer: Begin with the standard definition or identity of *sensors Explain the robot's machine vision.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain system Applications of robot vision systems. 3

Understanding M3.04 Sensors: Requirements of a sensor; Principles and Applications of the following types of sensors: Position sensors (Encoders, Resolvers, Piezo Electric); Range sensors; Proximity sensing; Force and torque sensing, Tactile sensors. Introduction to Machine Vision: Robot vision system (scanning and digitizing image data); Cameras (Acquisition of images); Vidicon camera (Working principle & construction); Applications of Robot vision system: Inspection, Identification, Navigation & serving.. (3 marks)

Answer: Begin with the standard definition or identity of *system Applications of robot vision systems. 3* *Understanding M3.04 Sensors: Requirements of a sensor; Principles and Applications of the following types of sensors: Position sensors (Encoders, Resolvers, Piezo Electric); Range sensors; Proximity sensing; Force and torque sensing, Tactile sensors. Introduction to Machine Vision: Robot vision system (scanning and digitizing image data); Cameras (Acquisition of images); Vidicon camera (Working principle & construction); Applications of Robot vision system: Inspection, Identification, Navigation & serving..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain 4 Understanding system Explain the advanced automation. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding system Explain the advanced automation.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain functions Explain the applications of robotics in. (3 marks)

Answer: Begin with the standard definition or identity of *functions* Explain the applications of robotics in. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding various industries. Automation: Basic elements of automated system, advanced automation functions, levels of automation. Industrial Applications: Application of robots in machining; welding; assembly, material handling, biomedical applications. Text / Reference: T/R Book Title/Author Mikell P. Weiss G.M., Nagel R.N., Odraj N.G., Industrial Robotics, McGraw R1 Hill Singapore, 1996 Ghosh, Control in Robotics and Automation: Sensor Based Integration, Allied R2 Publishers, Chennai, 1998. Saeed B. Niku, - Introduction to Robotics: Analysis, Systems, Applications - R3 Pearson Education Inc. New Delhi 2006. M.P. Groover, - Industrial Robotics: Technology, Programming and R4 Applications -Tata McGraw Hill Co, 2001. Fu.K.S. Gonzalz.R.C and Lee C.S.G - Robotics Control, Sensing, Vision and R5 Intelligence -, McGraw Hill Book Co, 1987. R6 Yoram Koren - Robotics for Engineers -McGraw Hill Book Co, 1992. Ganesh S. Hedge, - A Text book on Industrial Robotics -Laxmi Publications R7 Pvt. Ltd., New Delhi, 2008. S.R. Deb & Sankha Deb - Robotics Technology and Flexible Automation - R8 Tata McGraw-Hill, 2010. Elements of Robotics Process Automation, Mukherjee, Khanna Publishing R9 House, Delhi, 2018 R10 Issac Asimov I Robot, Ballantine Books, New York, 1986. Thomas R. Kurfess Ph.D., P.E. Robotics And Automation Handbook CRC R11 press Online Resources: SI.No Website Link 1 <https://nptel.ac.in/courses/112/101/112101098/> 2 <http://www.nptelvideos.in/2012/12/robotics.html> 3 <https://www.guru99.com/robotic-process-automation-tutorial.html>. (3 marks)

Answer: Begin with the standard definition or identity of 3 Understanding various industries. Automation: Basic elements of automated system, advanced automation functions, levels of automation. Industrial Applications: Application of robots in machining; welding; assembly, material handling, biomedical applications. Text / Reference: T/R Book Title/Author Mikell P. Weiss G.M., Nagel R.N., Odraj N.G., Industrial Robotics, McGraw R1 Hill Singapore, 1996 Ghosh, Control in Robotics and Automation: Sensor Based Integration, Allied R2 Publishers, Chennai, 1998. Saeed B. Niku, - Introduction to Robotics: Analysis, Systems, Applications - R3 Pearson Education Inc. New Delhi 2006. M.P. Groover, - Industrial Robotics: Technology, Programming and R4

Applications –Tata McGraw Hill Co, 2001. Fu.K.S. Gonzalz.R.C and Lee C.S.G - Robotics Control, Sensing, Vision and R5 Intelligence –, McGraw Hill Book Co, 1987. R6 Yoram Koren - Robotics for Engineers –McGraw Hill Book Co, 1992. Ganesh S. Hedge, - A Text book on Industrial Robotics –Laxmi Publications R7 Pvt. Ltd., New Delhi, 2008. S.R. Deb & Sankha Deb - Robotics Technology and Flexible Automation – R8 Tata McGraw-Hill, 2010. Elements of Robotics Process Automation, Mukherjee, Khanna Publishing R9 House, Delhi, 2018 R10 Issac Asimov I Robot, Ballantine Books, New York, 1986. Thomas R. Kurfess Ph.D., P.E. Robotics And Automation Handbook CRC R11 press Online Resources: Sl.No Website Link 1 <https://nptel.ac.in/courses/112/101/112101098/> 2 <http://www.nptelvideos.in/2012/12/robotics.html> 3 <https://www.guru99.com/robotic-process-automation-tutorial.html>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding laws of robotics..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Explain different Robotic Drive Systems 5 Understanding Explain different actuation methods 4 Understanding M2.02 Conversion between linear and rotary.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain the requirements of sensors 4 Understanding Explain the application of different.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **4 Understanding system Explain the advanced automation.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://nptel.ac.in/courses/112/101/112101098/>
- <http://www.nptelvideos.in/2012/12/robotics.html>
- <https://www.guru99.com/robotic-process-automation-tutorial.html>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 5083C. Verify institutional updates against the current official curriculum.